

Adaptation Utility Reveals Why Qualified Open-World Recognition Stages Fail to Compose

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Abstract

Open-world recognition is usually evaluated as a sequence of separate problems: reject unfamiliar inputs, cluster rejected examples, request labels, add classes, and route future inputs. Strong stage-level results do not guarantee that these stages compose into a useful system. We study this gap in a preregistered, human-review-budgeted loop over frozen 384-dimensional DINOv2 features. The protocol uses one primary endpoint: post-integration balanced accuracy on known and discovered classes at 50 reviewed labels per episode, with known-class regression and remaining-unknown recall as safety constraints.

The parent study first established that several stages could pass in isolation. A rank-32 class-conditional Gaussian achieved 92.08% known coverage, 86.17% unknown recall, 0.9561 AUROC, and 98.67% review precision. HDBSCAN achieved 100% group recall and precision in its stage-wise evaluation, and confirmed rank-16 class insertion improved isolated target success by 43.33 percentage points. Nevertheless, grouped end-to-end loops integrated 0 of 3 confirmable classes.

We then optimize review selection for adaptation utility rather than purity. Across nine seed-by-arrival cells, deterministic utility selection beats density-core selection in all nine cells by 3.593 balanced-accuracy points on average, with a paired 95% bootstrap interval of [2.794, 4.382]. This is a real and consistent signal, but it misses the preregistered 5-point minimum and violates the 2-point remaining-unknown-recall band in six cells. A separate local-residual study preserves 100% of unaffected predictions, repairing an earlier scope-leakage failure, but adds at most 0.204 points of utility and also fails unknown-recall safety. The final registered outcome is therefore negative: purity-selected discovery outputs are insufficient statistics for safe adaptation, while improving representativeness helps but does not by itself qualify an open-world lifecycle. The study contributes an auditable episode protocol, typed stage interfaces, fail-closed lineage contracts, exact rollback, and artifact-only reproduction of the negative result.

Keywords: open-world recognition, open-set recognition, novel-class discovery, human-in-the-loop learning, continual learning, adaptation utility, transactional machine learning, negative results

1 Introduction

A deployed classifier eventually encounters examples outside its original class inventory. Open-world recognition requires the system to recognize known classes, reject unknowns, and incrementally add newly supervised classes [1]. In an operational system, however, rejection is only the first step. Rejected evidence must be buffered; recurring structure must be separated from corruption; candidate groups must be reviewed; semantic labels must be supplied by an authorized actor; updates must be validated; and every publication must remain auditable and reversible.

The literature contains mature mechanisms for individual parts of this loop. OpenMax and the Extreme Value Machine calibrate unfamiliarity [2, 3]. Stream-mining systems buffer and organize emerging classes [4, 5, 6, 7, 8]. Generalized and continual category-discovery methods identify novel groups in modern representation spaces [9, 10, 11, 12]. Expert Gate and modular model systems route among specialists [13, 14, 15]. Yet independently successful stages may emit the wrong sufficient statistics for the next stage.

This paper asks:

Can a review-budgeted discovery, confirmation, adaptation, and routing loop improve post-integration performance while preserving known-class accuracy, remaining-unknown recall, and transactional safety?

We investigate this question in GEODE, an experimental framework for explicit geometric models and auditable model lifecycles. We do not assume that signed-distance geometry is necessary or competitively optimal. In the same frozen representation, an earlier weighted affine head reached 91.73% balanced accuracy, compared with 96.77% for an RBF SVM. The acceptance mechanism is therefore treated as substitutable, and a low-rank Gaussian is the retained primary head.

Our contributions are:

1. **A registered end-to-end endpoint.** We evaluate adaptation utility at a fixed human-review budget instead of promoting systems through purity, AUROC, or coverage alone.
2. **A quantified interface failure.** Dense core sets cover only 34.0–39.1% of the rejected region under a nearest-neighbor support diagnostic, whereas boundary-inclusive sets cover 99.37–99.78%.
3. **A consistent but insufficient selection signal.** Utility selection improves over core selection in 9/9 cells, but the gain is smaller than registered and does not preserve remaining-unknown recall.
4. **A transactional harness.** Typed interface contracts, immutable lineage, human-confirmation gates, exact rollback, deterministic replay, and artifact-only conclusion verification separate predictive failure from systems failure.

The central result is negative but actionable. Optimizing discovery for review purity does not provide sufficient support for adaptation. Optimizing support diversity improves adaptation, but safe open-world learning remains a joint constrained problem rather than a sequence of independently optimized stages.

2 Related Work

2.1 Open-set and open-world recognition

Bendale and Boulton formalized open-world recognition as known-class classification with explicit rejection and incremental class addition [1]. OpenMax calibrates activation vectors using extreme-value theory [2], while the Extreme Value Machine models class-conditional margin tails and supports incremental updates [3]. Feature-space alternatives include maximum posterior probability [17], Mahalanobis or class-conditional density scores [18, 19], deep nearest neighbors [20], and distance-aware discriminative models [21, 22].

These methods mainly answer whether an input lies outside learned support. Our primary question starts after rejection: whether selected rejected examples contain enough information to create a useful and safe child model.

2.2 Streaming novel-class detection

Data-stream systems operationalized reject-buffer-group-update loops before the modern deep-representation era. ECSMiner detects novel classes under concept drift [4]; SAND and ECHO address limited labels and evolving concepts [5, 6]; MINAS maintains micro-clusters and recurring novelty [7]; and SENCForest targets emerging classes [8]. These systems motivate our persistent buffer and delayed-confirmation protocol. We add immutable transaction lineage and measure downstream adaptation utility directly.

2.3 Category discovery

Generalized category discovery clusters unlabeled data containing both known and unknown categories, often over strong self-supervised features [9]. ORCA jointly addresses seen and novel classes in open-world semi-supervised learning [10]. Grow and Merge and incremental GCD extend discovery across time [11, 12]. Our study holds the DINOv2 representation [30] fixed so that review-set selection, rather than representation training, is the manipulated variable.

DBSCAN [23], HDBSCAN [24], and FINCH [25] offer distinct grouping mechanisms. Cluster quality is usually measured through coherence, purity, or recovery. We show that a pure cluster core can still be an inadequate adaptation set.

2.4 Expert routing and explicit models

Expert Gate uses reconstruction error to route among task experts [13]. More recent modular systems compose or route among independently trained experts [14, 15, 16]. Our protocol distinguishes compatibility from empirical support. A model fingerprint determines whether a bundle may be called; a learned support profile may only propose a shortlist. Stale or low-confidence routes fall back to exhaustive compatible-model evaluation.

Explicit geometry offers inspectable and editable artifacts [26, 27], while low-rank Gaussian class models are related to probabilistic PCA and mixtures of factor analyzers [28, 29]. Editability is valuable for governance, but does not imply predictive superiority. We therefore report geometric heads as ablations rather than premises.

3 Problem Formulation

Let $f(x) \in \mathbb{R}^d$ be a frozen representation. Before episode e , the parent model recognizes class inventory $\mathcal{K}_e$. Review and confirmation may add a discovered class set $\mathcal{D}_e$. The registered episode utility is

$$U_e = \text{BA}_{\mathcal{K}_e \cup \mathcal{D}_e}(\text{child}) - \text{BA}_{\mathcal{K}_e \cup \mathcal{D}_e}(\text{parent}). \quad (1)$$

Rejected or unavailable discovered labels count as incorrect for the parent. This prevents a no-update system from appearing competitive by excluding the new class from evaluation.

3.1 Fixed resources and safety

Each episode receives exactly $B = 50$ reviewed labels. Unspent labels do not transfer. The representation and episode partitions are immutable. A candidate must keep known-class regression below one percentage point and preserve remaining-unknown recall within two points of its parent. Semantic publication requires linked human confirmation. Replay, graph validation, rollback, lineage, and fallback checks must all pass.

3.2 Statistical advancement

The pivotal review-selection hypothesis is preregistered. Utility-selected review advances only if it:

1. exceeds density-core selection by at least 5.0 balanced-accuracy points in mean U_e ;
2. has a paired 95% bootstrap lower bound above zero;
3. improves at least 7 of 9 seed-by-episode cells; and
4. satisfies the review-budget, safety, and transaction contracts.

The first criterion is an effect-size requirement, not a significance test. Passing the interval and cell-count criteria is insufficient if the practical or safety thresholds fail.

4 System and Protocol

4.1 Data, representation, and episodes

The realized study is a bounded CIFAR-10 proxy using immutable 384-dimensional CLS-token features from DINOv2-small. Each frozen seed has 10,000 training and 1,000 development examples. Seeds are 11, 23, and 37. The v8 replay begins with classes 0–5 and introduces classes 6, 7, and 8 in order, yielding nine seed-by-arrival cells. Class 9 remains unknown. No untouched final-label schedule is opened.

Each class is deterministically partitioned into geometry fitting, anchor, review-candidate, and proxy-validation subsets. The validation subset may score a frozen candidate review set but never contributes adaptation support. Every episode records class order, partition hashes, support IDs, confirmation IDs, thresholds, parent and child hashes, utility, safety metrics, and rollback evidence.

4.2 Stage-interface contracts

We register five adjacent interfaces:

1. rejector $\rightarrow$ buffer;
2. buffer $\rightarrow$ clusterer;
3. clusterer $\rightarrow$ review selector;
4. review $\rightarrow$ adapter; and
5. adapter $\rightarrow$ router.

Seven versioned schemas represent the utility endpoint, episode replay, interface audits, threshold transfer, review selection, local residual scope, and integrated routing. The harness aborts on partition leakage, stale class order, threshold-lineage mismatch, review-budget overflow, missing confirmation, or rollback-parent drift.

4.3 Acceptance and threshold transfer

The retained acceptance head is a rank-32 class-conditional Gaussian. For center μ , orthonormal tangent basis V , tangent variances λ_j , and residual variance σ^2 ,

$$q(x) = \sum_j \frac{(v_j^\top (x - \mu))^2}{\lambda_j} + \frac{\|(I - VV^\top)(x - \mu)\|_2^2}{\sigma^2}. \quad (2)$$

The class log likelihood includes the corresponding log determinant, and novelty is $-\max_k \log p(x \mid k)$. We compare a frozen parent threshold, a class-count heuristic, a global anchor quantile, and per-class anchor thresholds with a global fail-closed ceiling. Stale anchor lineage restores the parent threshold.

4.4 Review-selection arms

All eligible arms receive the same candidate pool and exactly 50 labels:

Core-selected. Samples nearest the candidate-group center.

Boundary-inclusive. A deterministic mix of core examples, modes, and parent-boundary-distance quantiles.

Coverage-selected. Farthest-first k -center selection.

Utility-selected. Deterministic candidate subsets are ranked by a cheap rank-3 proxy adapter on a disjoint validation partition; the production adapter remains rank 16.

Random stratified. Seeded sampling across two candidate modes.

Review-everything. All 300 candidate examples; diagnostic only and ineligible for equal-budget advancement.

All selected samples belong to the arriving class in this controlled sufficient-statistics study. Purity is therefore fixed at 100%; the experiment isolates representativeness.

4.5 Transactional adaptation and routing

The production adapter adds one confirmed rank-16 low-rank component and recalibrates the global anchor threshold. Each transaction appends exactly one class, links a persistent review and confirmation, and must restore the exact parent bundle under rollback. Routing remains exhaustive because no empirical router passed the parent study’s authoritative gate.

Table 1: Parent stage-wise results. Values are three-seed means.

Stage	Retained mechanism	Result
Rejection	Rank-32 Gaussian	86.17% unknown recall 92.08% known coverage 98.67% review precision
Discovery	HDBSCAN	100% group recall/precision
Discovery	FINCH	83.33% recall; 86.39% precision
Adaptation	Confirmed rank-16 insertion	+43.33 points target success
Routing	Best shadow profile	99.57% winner inclusion

Table 2: Threshold transfer over three development seeds.

Rule	Mean known drop	Mean unknown drop	Pass
Frozen parent	0.000	0.000	Yes
Class-count heuristic	0.000	0.000	Yes
Global anchor quantile	0.000	0.000	Yes
Per-class anchor	1.875	-8.667	No

4.6 Local residual branch

Two preregistered residuals are evaluated independently after the pivotal selection experiment: a local target-class temperature and a bounded class-local affine correction. Before fitting, the parent alone freezes the affected region using selected support, a bounded nearest-neighbor radius, and component responsibility. Residual fusion is applied only inside that mask. An attempt requires at least 5 points of utility improvement, 99.9% unaffected-prediction preservation on every seed, and the same known and unknown safety bounds.

5 Results

5.1 Parent stage-wise evidence

Table 1 summarizes the stage-level mechanisms inherited from the v7 parent study. These results motivated composition but did not establish it.

Despite these stage-level results, HDBSCAN and FINCH integrated 0/3 confirmable classes in the v7 grouped loop. Reviewing every rejection integrated 3/3 but provided no review reduction. This composition failure motivated the v8 endpoint.

5.2 Threshold transfer

Frozen, class-count, and global anchor-quantile thresholds all satisfy the registered bands on seeds 11, 23, and 37. We retain the global anchor rule because it directly recalibrates after class-order change. Mean known-class regression and mean remaining-unknown-recall change are both 0.00 points; mean remaining-unknown recall is 79.33%. Per-class fail-closed thresholds increase unknown rejection but cause 1.875 points of mean known regression, exceeding the one-point ceiling.

5.3 Core sets are geometrically unrepresentative

Table 3 compares 50 core examples with 50 boundary-inclusive examples. Both cover the two coarse k -means modes, so mode count alone does not expose the defect. Nearest-neighbor support and covariance do. Core sets cover only 34.0–39.1% of the rejected candidate region; boundary-inclusive sets cover more than 99% and approximately recover the full covariance trace.

In the M46 proxy adapter, boundary inclusion adds 0.259 balanced-accuracy points over core selection on average. The effect is positive on two of three seeds but is much smaller than the pivotal M47 threshold.

Table 3: Review-set representativeness diagnostics.

Seed	Core NN cov.	Boundary NN cov.	Core cov. ratio	Boundary cov. ratio
11	34.04%	99.68%	0.663	1.078
23	35.47%	99.78%	0.658	1.096
37	39.14%	99.37%	0.679	1.068

Table 4: Mean adaptation utility and review cost across nine cells.

Review policy	Mean U_e	Mean labels	Mean remaining-unknown recall
Core-selected	6.36	50	59.72%
Boundary-inclusive	9.71	50	75.98%
Coverage-selected	10.62	50	82.50%
Utility-selected	9.96	50	78.15%
Random stratified	9.70	50	75.41%
Review-everything	11.26	300	86.13%

The interface audit ranks changed-region fusion scope, full normalized router scores, and boundary examples as the three measured mismatches.

5.4 Equal-budget adaptation utility

Table 4 reports mean episode utility across nine cells. Coverage selection is the strongest eligible deterministic arm at 10.62 points. Utility selection reaches 9.96 points, compared with 6.36 for core selection. Review-everything reaches 11.26 points but uses 300 labels per episode and is ineligible.

Utility selection beats core selection in 9/9 cells. The mean paired difference is 3.593 points, and the 10,000-resample paired percentile bootstrap interval is [2.794, 4.382]. Thus the sign is stable and the interval excludes zero. Nevertheless, the gain is below the registered 5-point practical threshold.

The safety conjunction also fails. Every utility-selected transaction consumes exactly 50 reviews, links confirmation, appends one class, replays, and rolls back exactly. Mean known regression is negative, indicating improvement rather than degradation. However, six of nine cells lose more than two points of remaining-unknown recall. The registered pivotal gate therefore fails and blocks learned selection and final E12 qualification.

5.5 Locality can be repaired without creating utility

The parent A3 study preserved only 84.2% of unaffected predictions because an ostensibly local edit entered a global fusion path. Parent-only region masks and explicit fusion gating raise unaffected preservation to 100% for both M49 attempts. Maximum known regression is only 0.125 points.

The residuals do not qualify. Local target temperature adds 0.000 points of mean utility. The bounded affine correction adds 0.204 points, far below the 5-point gate. Both observe a maximum remaining-unknown-recall drop of 4.333 points. No residual is retained.

6 Discussion

6.1 Purity is not an adaptation endpoint

The parent study appears strong when read stage by stage: high unknown recall, near-perfect review precision, perfect HDBSCAN recovery, and large isolated new-class gains. The integrated failure and v8 selection results show why this reading is misleading. A dense core can be semantically pure while discarding low-density modes and boundary variation needed by the child model.

Table 5: Preregistered pivotal gate.

Operand	Observed	Pass
Mean gain over core ≥ 5.0 points	3.593	No
Paired 95% lower bound > 0	2.794	Yes
Positive cells $\geq 7/9$	9/9	Yes
Budget, safety, and transaction conjunction	6 unknown-recall violations	No

The 9/9 utility gain over core selection is evidence for the sufficient- statistics hypothesis. The failure of the 5-point and safety gates is equally important. Representativeness is necessary in this harness, but not sufficient. Future objectives must jointly optimize adaptation utility and preservation of the remaining unknown space.

6.2 Why the proxy selector is not the winner

Coverage-selected review outperforms utility-selected review. The utility selector ranks subsets with a cheap rank-3 proxy but deploys a rank-16 production adapter. This creates proxy regret: validation utility under the cheap model does not perfectly rank production outcomes. Training a learned selector might reduce regret, but the preregistered kill switch correctly prohibits that continuation after the pivotal endpoint fails.

6.3 Unknown recall is a lifecycle state variable

Adding a class changes the score competition and the meaning of a calibrated threshold. Global anchor recalibration transfers safely in the isolated M46 experiment, yet six M47 cells still exceed the remaining-unknown band. This shows that threshold transfer cannot be evaluated independently from support selection and cumulative class addition. Unknown recall must be measured after each transaction, not inherited from the rejection benchmark.

6.4 Locality and rollback answer different questions

Rollback proves that the parent can be restored. It does not prove that the child update is local. The M49 result demonstrates a direct repair: parent-defined masks plus explicitly gated fusion preserve every unaffected prediction. The absence of utility gain prevents a stronger claim, but the mechanism clarifies how edit scope should be enforced.

6.5 Negative results and stopping rules

The final outcome is not “no signal.” Utility selection has a positive, consistent effect. The outcome is negative because the effect is smaller than the registered practical threshold and violates a safety co-primary. Publishing only the positive interval would overstate the evidence. The stopping rule prevents post-hoc learner tuning, head substitution, or untouched schedule access from converting a failed primary endpoint into a new claim.

7 Reproducibility

All milestone configurations, schemas, selected sample IDs, parent and child lineage, and result artifacts are stored in the project repository. The final artifact-only command is:

```
.\.venv\Scripts\python.exe -m experiments.tier1.eval_v8_final_replay
```

The verifier checks four immutable milestone indexes, 11 indexed artifacts, and six conclusion operands. Two runs must be byte-identical. The verifier loads no training data and opens no final labels. The complete repository test gate contains 374 tests.

For an archival release, the authors should insert the public repository URL and release DOI here: repository URL placeholder. The arXiv source package should include this `.tex` file; no generated evidence or benchmark data are required to compile the manuscript.

8 Limitations

Proxy scale. The realized study uses CIFAR-10 classes and frozen DINOv2 features. It does not establish performance on large semantic inventories, natural long-running streams, or domain-scale routing.

Development-only evidence. The untouched M50 schedule remains sealed because the M47 kill switch fired. The result is preregistered development evidence, not an independent final-test confirmation.

Simulated review. A deterministic oracle models human confirmation. Real reviewers introduce disagreement, delay, expertise variation, and context-dependent cost.

Controlled candidate purity. The v8 sufficient-statistics experiment uses candidates from the true arriving class to isolate review-set geometry. Real discovery also introduces false members and fragmented groups.

Limited selectors and adapters. The deterministic utility selector enumerates registered subset families and uses a rank-3 proxy. The result does not rule out better constrained selectors, but the registered program prohibits training them after the pivotal failure.

No head-substitution conclusion. The same-harness geometric and kNN substitutions are not run after the binding kill switch. Earlier stage-level comparisons cannot be relabeled as adaptation-utility evidence.

No universal novelty claim. The prior-art audit cannot rule out patents, non-English work, inaccessible publications, or closed industrial systems.

9 Ethical and Deployment Considerations

Open-world systems can assign semantic meaning to coherent but irrelevant, sensitive, or harmful clusters. This study prohibits autonomous semantic publication and requires linked confirmation, immutable provenance, and exact rollback. These controls reduce but do not eliminate risk. Real deployment should additionally support reviewer disagreement, appeal and deletion procedures, privacy-aware retention, subgroup performance audits, monitoring for feedback loops, and explicit authority boundaries.

The experiments use public benchmark representations and simulated review. They do not involve human subjects or sensitive personal data. The architecture should not be interpreted as authorization for high-stakes use.

10 Conclusion

We evaluated an open-world learning loop through adaptation utility rather than stage-wise proxies. Strong rejection, grouping, and isolated adaptation results did not compose. Review sets selected for density-core purity were geometrically unrepresentative. Boundary- and coverage-aware selection improved post-integration balanced accuracy consistently, but the registered utility-selected arm gained only 3.593 points rather than 5 and failed the remaining-unknown safety constraint.

The broader methodological lesson is that open-world recognition should be evaluated as a constrained lifecycle. Rejection recall, cluster purity, adaptation accuracy, routing inclusion, locality, and rollback can all look strong independently while the complete system remains unqualified. The interfaces between stages—and the information each consumer actually needs—must be treated as first-class experimental objects.

References

- [1] A. Bendale and T. Boulton. Towards open world recognition. In *CVPR*, 2015.
- [2] A. Bendale and T. Boulton. Towards open set deep networks. In *CVPR*, 2016.
- [3] E. Rudd, L. Jain, W. Scheirer, and T. Boulton. The extreme value machine. *IEEE TPAMI*, 2018.
- [4] M. Masud et al. Classification and novel class detection in concept-drifting data streams under time constraints. *IEEE TKDE*, 2011.
- [5] A. Haque et al. SAND: Semi-supervised adaptive novel class detection and classification over data stream. In *AAAI*, 2016.
- [6] A. Haque et al. ECHO: A data stream classification method for detecting novel class drift. In *ICDE*, 2016.
- [7] E. de Faria, A. Gama, and A. Carvalho. Novelty detection algorithm for data streams multi-class problems. *Data Mining and Knowledge Discovery*, 2016.
- [8] X. Mu, F. Zhu, J. Du, E.-P. Lim, and Z.-H. Zhou. Classification under streaming emerging new classes. *IEEE TKDE*, 2017.
- [9] S. Vaze et al. Generalized category discovery. In *CVPR*, 2022.
- [10] K. Cao, M. Brbić, and J. Leskovec. Open-world semi-supervised learning. In *ICLR*, 2022.
- [11] Z. Zhang et al. Grow and Merge: A unified framework for continuous categories discovery. In *NeurIPS*, 2022.
- [12] B. Zhao and O. Mac Aodha. Incremental generalized category discovery. In *ICCV*, 2023.
- [13] R. Aljundi, P. Chakravarty, and T. Tuytelaars. Expert Gate: Lifelong learning with a network of experts. In *CVPR*, 2017.
- [14] S. Gururangan et al. DEMix layers: Disentangling domains for modular language modeling. arXiv:2108.05036, 2021.
- [15] M. Li et al. Branch-Train-Merge: Embarrassingly parallel training of expert language models. arXiv:2208.03306, 2022.
- [16] C. Huang et al. LoraHub: Efficient cross-task generalization via dynamic LoRA composition. In *COLM*, 2024.
- [17] D. Hendrycks and K. Gimpel. A baseline for detecting misclassified and out-of-distribution examples in neural networks. In *ICLR*, 2017.
- [18] K. Lee et al. A simple unified framework for detecting out-of-distribution samples and adversarial attacks. In *NeurIPS*, 2018.
- [19] J. Mukhoti et al. Deep deterministic uncertainty: A new simple baseline. *TMLR*, 2023.
- [20] Y. Sun et al. Out-of-distribution detection with deep nearest neighbors. In *ICML*, 2022.
- [21] J. van Amersfoort et al. Uncertainty estimation using a single deep deterministic neural network. In *ICML*, 2020.
- [22] J. Liu et al. Simple and principled uncertainty estimation with deterministic deep learning via distance awareness. In *NeurIPS*, 2020.
- [23] M. Ester, H.-P. Kriegel, J. Sander, and X. Xu. A density-based algorithm for discovering clusters in large spatial databases with noise. In *KDD*, 1996.

- [24] R. Campello, D. Moulavi, and J. Sander. Density-based clustering based on hierarchical density estimates. In *PAKDD*, 2013.
- [25] S. Sarfraz, V. Sharma, and R. Stiefelhausen. Efficient parameter-free clustering using first neighbor relations. In *CVPR*, 2019.
- [26] A. Requicha. Representations for rigid solids: Theory, methods, and systems. *ACM Computing Surveys*, 1980.
- [27] J. Park et al. DeepSDF: Learning continuous signed distance functions for shape representation. In *CVPR*, 2019.
- [28] M. Tipping and C. Bishop. Probabilistic principal component analysis. *JRSS B*, 1999.
- [29] Z. Ghahramani and G. Hinton. The EM algorithm for mixtures of factor analyzers. Technical Report CRG-TR-96-1, University of Toronto, 1996.
- [30] M. Oquab et al. DINOv2: Learning robust visual features without supervision. *TMLR*, 2024.

A Final Branch Disposition

Table 6: Registered v8 milestone disposition.

Milestone	Decision	Final status
M45	Endpoint, schemas, replay	Passed
M46	Threshold and interface audit	Passed
M47	Equal-budget utility selection	Outcome D
M48	Learned selection	Blocked by M47
M49	Two local-residual attempts	No residual retained
M50/E12	Untouched lifecycle qualification	Blocked by M47

B Artifact Verification

The final replay reports:

- four verified milestone indexes;
- 11 verified indexed artifacts;
- six reproduced conclusion operands;
- byte-identical output across two executions;
- no training-data loading; and
- no final-label access.